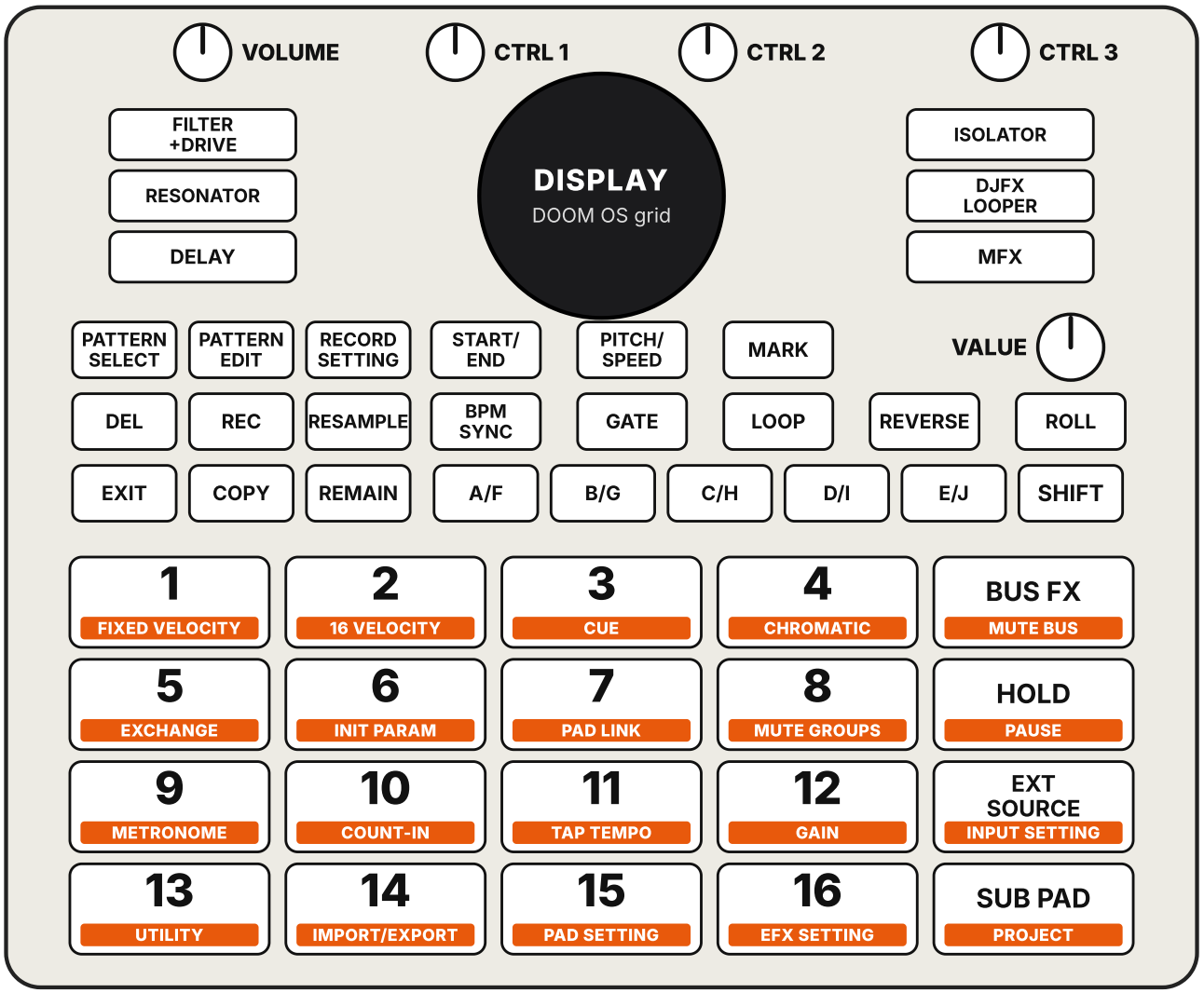
DOOM OS · SP-404MKII

Pocket cheat sheet & panel map – a graphical sequencer layer for the Roland SP-404MKII



HOLD this key · PRESS / TAP this key · TURN this knob · 1 order of steps
orange bar = Roland's printed SHIFT function for that pad

WHERE THINGS LIVE

- **Display + FX row** – DOOM OS draws its grid on the display. FX buttons (FILTER+DRIVE, RESONATOR, DELAY, ISOLATOR, DJFX LOOPER, MFX) flank it.
- **Knobs** – CTRL 1 / 2 / 3 edit whatever the screen shows; VALUE scrolls (turn) and confirms (press).
- **Left block** – PATTERN SELECT, PATTERN EDIT, RECORD SETTING / DEL, REC, RESAMPLE / EXIT, COPY, REMAIN.
- **Right block** – START/END, PITCH/SPEED, MARK / BPM SYNC, GATE, LOOP, REVERSE, ROLL / bank buttons A/F–E/J and SHIFT.
- **Pads** – 16 pads in a 4×4 grid, numbered left→right, top→bottom. In DOOM OS they are the **16 steps**. BUS FX, HOLD, EXT SOURCE and SUB PAD sit in the column on the right.

GOLDEN RULES

- **Open**: hold REMAIN + press SUB PAD. **Close**: EXIT (saves).
- **Lost?** Hold SHIFT for 3 s, then press any control to see what it does. EXIT leaves help.
- EXIT always backs out one level: closes menus, cancels pattern tools, bounces and parameter-lock edits.
- **Undo / redo**: SHIFT + PATTERN SELECT / REMAIN + PATTERN SELECT.
- **Saving is automatic** after edits, on pattern change and on exit.
- Combos are written **hold + press**: the first key is held down, the second is tapped.

BOOKLET MAP

p2 Getting around
open · play · patterns · tempo · tracks · steps · menus · grid · piano roll

p3 Record & edit
rec modes · quantize · bounce · pattern tools · parameter locks

p4 Sound & mix
instruments · kits · arp · mixer · EQ · side chain

p5 FX & reference
FX · automation · settings · master shortcut table

Unofficial cheat sheet compiled from the DOOM OS 0.7 manual (Klangfeld Labs) with panel layout taken from Roland's SP-404MK2 Reference Manual. Not affiliated with Roland. Diagrams are schematic, not to scale.

Getting around

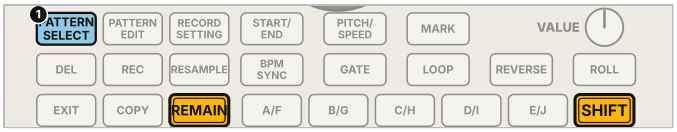
Open · play · patterns · tempo · tracks · steps · menus · grid · piano roll

Open / close DOOM OS



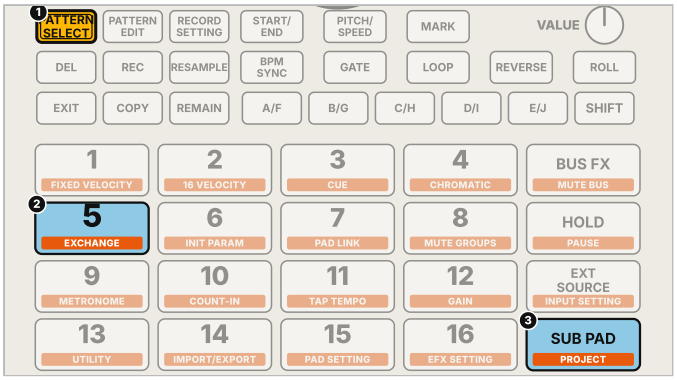
Open: hold **REMAIN** + press **SUB PAD**.
Close: press **EXIT** – saves changes. A *Hold to Exit* setting can require holding it.

Play / stop · undo · redo



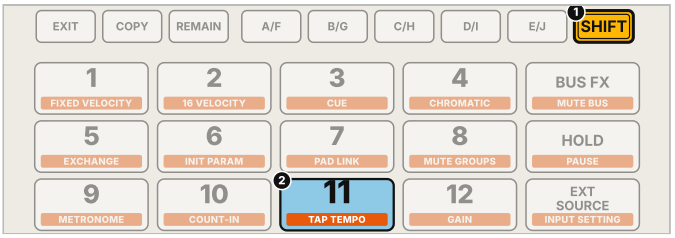
Play / stop: **PATTERN SELECT**
Undo: hold **SHIFT** + **PATTERN SELECT**
Redo: hold **REMAIN** + **PATTERN SELECT**

New / switch pattern



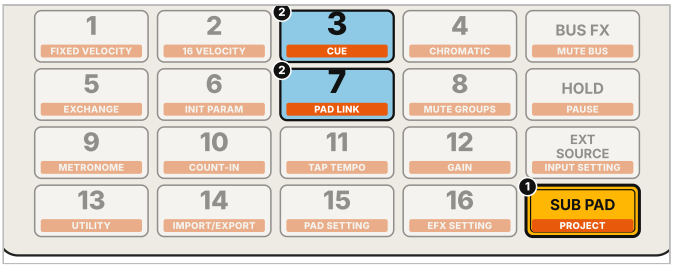
New: hold **PATTERN SELECT** + an **empty** pad.
Switch: hold **PATTERN SELECT** + a pad – starts when the playing pattern ends. **Add** **SUB PAD** to switch immediately.

Tempo



SHIFT + **pad 11** (TAP TEMPO), or press **RECORD SETTING** twice > BPM.

Add a track



Sample track: hold **SUB PAD**, tap sample pads, release.
Instrument track: hold **SUB PAD** + tap an **empty** pad (opens the instrument browser, p4).

Steps & notes (CTRL 1 / 2 / 3)



The 16 pads are the 16 **steps**: tap to toggle, hold a pad to select that note.
CTRL 1 velocity · **CTRL 2** length · **CTRL 3** nudge. Turn with **no pad held** = whole track; hold a pad = that one note.

Do this	Input	Result
Fine edits	**SHIFT** + **CTRL 2** · **SHIFT** + **CTRL 1**	Length in 1/128 steps · pitch up to ±12 semitones. Keep a selection: hold pad + **REMAIN**
Mute / solo track	**SHIFT** + press **VALUE** · **SHIFT** + **ROLL**	
Copy / paste / delete	**COPY**, then **RECORD SETTING** > EDIT > PASTE · **DEL**	DEL asks first; CLEAR... asks YES/NO; PASTE does not ask
TRACK page	**RECORD SETTING** once	EDIT (SWAP PAD, PASTE, CLEAR..., BOUNCE...) · PATTERN (CLEAR..., BOUNCE...) · GENERATE (PRINT ARP..., CLEAR...) · VIEWS (MIXER, AUTOMATION, ARP, SIDECHAIN)
PATTERN panel	**RECORD SETTING** twice (3rd closes)	BPM · LENGTH (bars) · REC mode

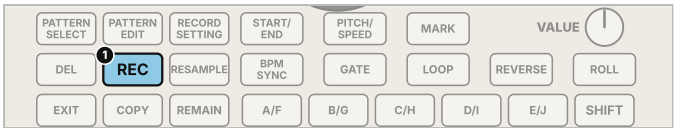
Patterns over 512 notes, and effect movements, can't be undone.

Unofficial cheat sheet compiled from the DOOM OS 0.7 manual (Klangfeld Labs) with panel layout taken from Roland's SP-404MK2 Reference Manual. Not affiliated with Roland. Diagrams are schematic, not to scale.

Record & edit

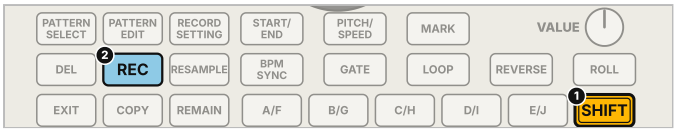
Recording modes · quantize · bounce in place · pattern tools · parameter locks

Record (REC)
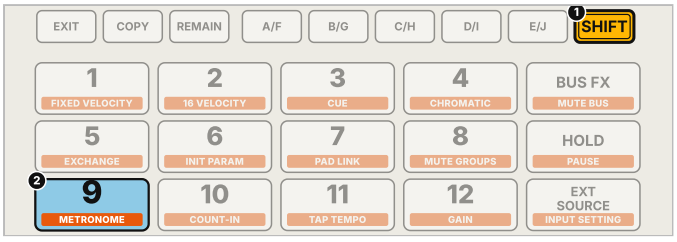


1st press = **rehearsal** (nothing recorded). 2nd press = **records**. **EXIT** stops recording (pattern keeps playing). **HOLD** is ignored while REC is on.

Record mode



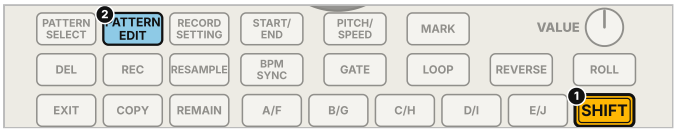
SHIFT + **REC** cycles **SINGLE** · **MULTI** · **CHROMATIC** (pad 5 = original pitch) · **16 VEL** (pad 4 = 127, pad 13 = 1). Also set in the PATTERN panel.

Metronome



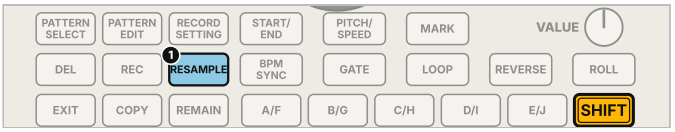
SHIFT + **pad 9** cycles ON · RECPLAY · OFF.

Input quantize (new notes)



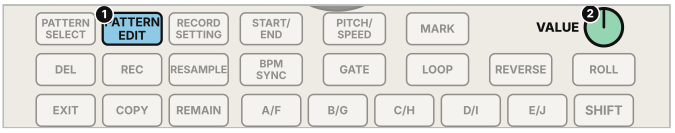
SHIFT + **PATTERN EDIT** sets **QNT** · **SWG** · **VEL** · **LEN** for newly recorded notes.

Bounce in place



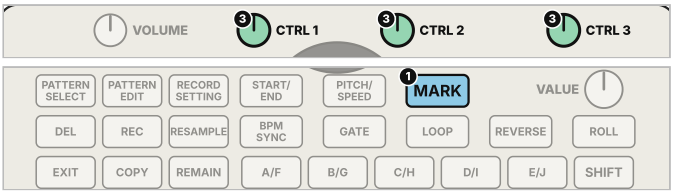
RESAMPLE = bounce the **track**; **SHIFT** + **RESAMPLE** = **everything you hear**. Pick type with **VALUE**, tap an **empty pad**, then press **RESAMPLE** or **VALUE**. A track bounce swaps the track's notes for one note on the new pad – **SHIFT** + **PATTERN SELECT** undoes it; **EXIT** cancels.

Pattern tools



PATTERN EDIT > turn **VALUE** to a tool > press **VALUE** > set amount > press **VALUE** to apply. Tap a track pad to limit to that track; tap again for ALL.

Parameter locks



MARK (grid or editor) > tap a step > edit with **CTRL 1–3** > **MARK** saves, **EXIT** discards. Locked pads show **blue**; **DEL** resets a step.

Pattern tool	Range / note
QUANT	1/4 to 1/32, incl. triplets
SWING	50–75; **SHIFT** switches 8th / 16th
SHIFT	±16 steps, wraps around
DOUBLE · HALVE	DOUBLE max 64 bars / 512 notes; HALVE not on a 1-bar pattern
REVERSE	–
HUMAN · TRANSP	HUMAN 1–30 · TRANSP ±12 semitones

Bounce type	What it captures
CUT	Pattern length
TAIL	Includes effect decay, up to 30 s
LOOP	Loops from the second pass

PARAMETER-LOCK FACTS

- Per-step instrument parameters, instrument tracks only.
- Locks follow notes when moved (SHIFT tool, QUANTIZE, DUPLICATE, CROP, SWAP PAD, COPY, PASTE).
- Limits: 4096 locks per project · 64 steps per session · 126 locks per track in a pattern.

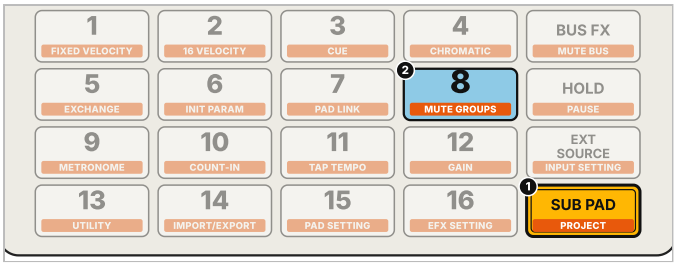
View	Input	Result
Grid views	**SHIFT** + **RECORD SETTING**	Cycles GRID (8 tracks) · LABELS · BEATS · BOTH · ZOOM (4 larger tracks)
Zoom · move · follow	**REMAIN** +turn **VALUE** · **SHIFT** +turn **VALUE** · **SHIFT** + **HOLD**	Zoom in ZOOM view · move by bar (beat at 1-beat zoom) · toggle Follow (playhead)

Unofficial cheat sheet compiled from the DOOM OS 0.7 manual (Klangfeld Labs) with panel layout taken from Roland's SP-404MK2 Reference Manual. Not affiliated with Roland. Diagrams are schematic, not to scale.

Sound & mix

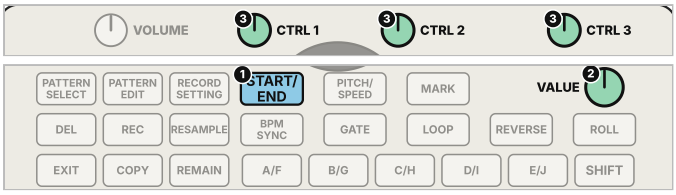
Instruments · drum kits · arpeggiator · mixer · EQ · side chain · waveform

Instrument browser



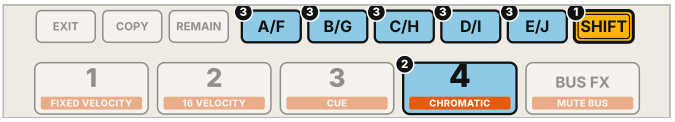
Hold **SUB PAD** + tap an **empty** pad, release. Turn **VALUE** to choose; hold **SUB PAD** to preview; **VALUE** loads. 20 instruments and 11 drum kits are built in.

Instrument editor



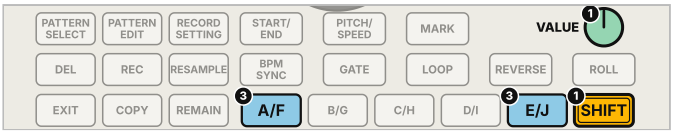
On an instrument track press **START/END**. Turn **VALUE** to pick a section (TONE, MOD, AMP, OUT…); **CTRL 1–3** edit; **SHIFT** = finer steps; **EXIT** saves.

Chromatic keys & octave



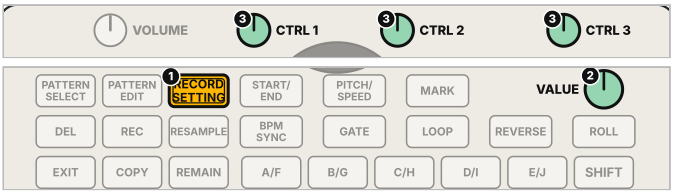
In the editor: **SHIFT** + **pad 4**. Octave via the BANK buttons: **C/H** default · **B/G**, **A/F** down · **D/I**, **E/J** up.

Drum kits



SHIFT + turn **VALUE** to KITS, choose, press **VALUE**, pick a bank with a BANK button, press **VALUE**. A non-empty bank asks to overwrite (can't undo); protected banks are refused.

Mixer



Hold **RECORD SETTING** 0.6 s. **VALUE** selects a track; **CTRL 1** pan · **CTRL 2** bus (B1 · B2 · DRY) · **CTRL 3** level. Tap a pad = mute; **REMAIN** + pad = solo. **VALUE** cycles pan / bus / level for fine adjust; **EXIT** deselects.

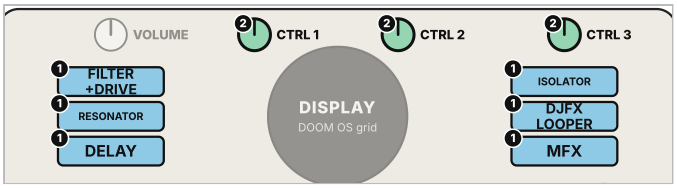
Tool	How	Details
Arpeggiator	**RECORD SETTING** > ARP, or the grid while playing	Hold pads to arp. **VALUE** toggles arp and swaps PLAY / PITCH pages; **HOLD** keeps it running. Persists until power-off. **PLAY**: **CTRL 1** TYPE (UP, DOWN, U/D, RND, ORD, CHRD, CONV, DIV, PNKY, THMB, U/D+) · **CTRL 2** RATE 1/4–1/32 + triplets · **CTRL 3** GATE 1–100%. **PITCH**: KEY (C–B) · SCALE (40) · OCT (±3). Print: **RECORD SETTING** > GENERATE > PRINT ARP… (replaces all notes on the track)
Track EQ	in mixer, **PITCH/SPEED**	**CTRL 1** gain ±12 dB · **CTRL 2** freq 20 Hz–20 kHz · **CTRL 3** Q 0.3–10. **VALUE** opens the screen; again = ADV band select. Up to 16 pads; EQ belongs to the pad
Side chain	**RECORD SETTING** > SIDECHAIN	**VALUE** on a **source** bus, then a **target** bus; **VALUE** twice on one bus = compress it. **CTRL 1** THR · **CTRL 2** RAT · **CTRL 3** REL; GAIN via **VALUE**. Off: hold **DEL** + **VALUE** on the bus. Shared with the unit's own side chain, all projects
Piano roll	press **VALUE** on the grid	Hold a step's pad + press another pad to set pitch (pad 5 = original; +1 semitone per pad across, and up the rows). Turn **VALUE** scrolls; **SHIFT**+turn changes track
Waveform display	**SHIFT** + **START/END**	1× output · 2× selected track · 3× hide. Shows the last two seconds
Replace instrument	in editor: **REMAIN**, choose, YES	Notes stay, parameters reset
Remove instrument	outside DOOM OS: hold **DEL** + pad, then **VALUE**	Cannot be undone
Mixer display	**SHIFT** + **RECORD SETTING**	ARROWS / DITHER, with or without NAMES

Unofficial cheat sheet compiled from the DOOM OS 0.7 manual (Klangfeld Labs) with panel layout taken from Roland's SP-404MK2 Reference Manual. Not affiliated with Roland. Diagrams are schematic, not to scale.

FX & quick reference

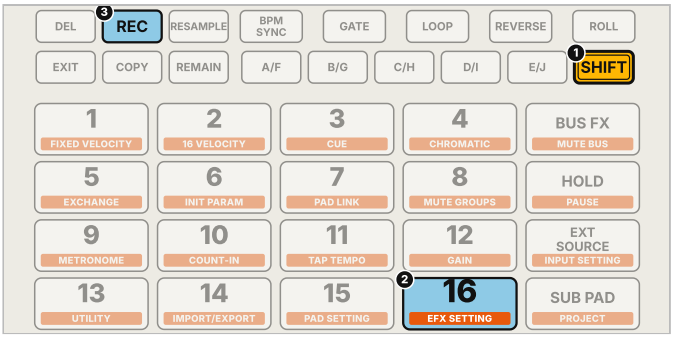
Effects · automation · settings · every shortcut on one page

Effects



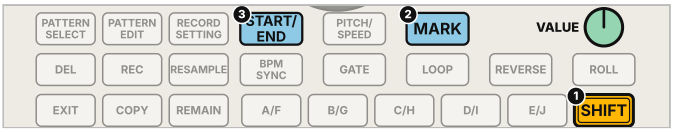
Each FX button turns its effect **on** and opens its page (stays open until **EXIT**; the effect stays on). Press again to turn **off**. **CTRL 1–3** set parameters; hold **VALUE** for page 2. **REMAIN** + FX button reopens a page without toggling.

Bus FX settings · record movements



SHIFT + **pad 16** opens bus effects settings. Record movements: open an FX page, press **REC** twice, turn knobs. **BUS FX** switches buses.

FX automation



SHIFT + **MARK** opens it. Turn **VALUE** for a parameter; **SHIFT** + turn = next effect. Press **VALUE** for per-step view, tap a pad, turn **VALUE**. **START/END** toggles **EFX MOTION PLAY** – movements only play while it is on.

MASTER SHORTCUT TABLE

Function	Input	Function	Input
Open DOOM OS	**REMAIN** + **SUB PAD**	Bounce track / mix	**RESAMPLE** / **SHIFT** + **RESAMPLE**
Undo / Redo	**SHIFT** + **PATTERN SELECT** / **REMAIN** + **PATTERN SELECT**	Instrument editor	**START/END**
Mixer	hold **RECORD SETTING** 0.6 s	Parameter locks	**MARK**
Help mode	hold **SHIFT** 3 s	FX automation	**SHIFT** + **MARK**
Metronome	**SHIFT** + **pad 9**	Waveform display	**SHIFT** + **START/END**
Record mode	**SHIFT** + **REC**	Bus FX settings	**SHIFT** + **pad 16**
Tempo	**SHIFT** + **pad 11**	UTILITY menu	**REMAIN** + **PATTERN EDIT**
MFX effect select	hold **MFX** + pad (17–32 via **DJFX LOOPER**; 33–48 via **ISOLATOR**)	Piano roll	press **VALUE** on the grid
Input quantize	**SHIFT** + **PATTERN EDIT**	DOOM OS settings	**EXIT**, then **SHIFT** + **pad 13** > SYSTEM > DOOM OS

PAD SHIFT FUNCTIONS

1 FIXED VELOCITY	**2** 16 VELOCITY	**3** CUE	**4** CHROMATIC
5 EXCHANGE	**6** INIT PARAM	**7** PAD LINK	**8** MUTE GROUPS
9 METRONOME	**10** COUNT-IN	**11** TAP TEMPO	**12** GAIN
13 UTILITY	**14** IMPORT/EXPORT	**15** PAD SETTING	**16** EFX SETTING

Bold = pad shortcuts DOOM OS refers to. Labels are Roland's printed SHIFT functions.

DOOM OS SETTINGS

Path: **EXIT**, then **SHIFT** + **pad 13** > SYSTEM > **DOOM OS** tab (**CTRL 3** navigates). Options: Mixer Level (ARROWS / DITHER) · Mixer Names · Boot Anim (GRAFFITI, VHS, HELLO, BUBBLE, SKETCH, HYPERSPACE, RANDOM) · Pattern View · Grid Lines · Pattern Follow · Motion Rec (OVERWRITE / OVERDUB) · Hold to Exit (0.1–1.0 s or OFF) · Boot to DOOM (YES / NO). The SHORTCUT tab holds up to 8 custom shortcuts.

Unofficial cheat sheet compiled from the DOOM OS 0.7 manual (Klangfeld Labs) with panel layout taken from Roland's SP-404MK2 Reference Manual. Not affiliated with Roland. Diagrams are schematic, not to scale.